Worked Example 1 — The Price-Plus-Burden Worksheet, filled in for an ambient-documentation purchase

A companion to Instrument 3 of The Operator's Thesis (Chapter 6), with Instruments 6 and 8. Hypothetical composite: the organization, the people, the vendor and every figure below are invented for teaching and are labeled illustrative. Nothing here describes a real deployment, a real vendor, or any employer, client, deal or fund. Version 1, October 2026.

The situation (illustrative)

Riverbend Health is a composite: a 300-bed community system with 140 clinicians in primary care and eleven specialties — 112 employed by Riverbend and 28 affiliated, in independent practices on its medical staff — a three percent operating margin, and a physician-retention problem that everyone in the building can name. An ambient-documentation vendor has proposed a two-wave rollout to all 140 clinicians. The invoice is the left-hand column of the spreadsheet and it is one line:

The invoice (left-hand column)	Illustrative
License: 140 clinicians × $400 per clinician per month × 12	**$672,000** in year one

The chief financial officer asked the question Chapter 6 says she would: what will this actually cost us? This sheet is the right-hand column — the bill the invoice never shows — filled in before the budget meeting, with a named role on every line and hours times loaded rates. The hours are estimates the project team wrote down and the vendor was asked to check; the rates are loaded internal rates (salary, benefits and overhead) rounded for the example.

The worksheet, filled in

The burden line (beyond the invoice)	Who carries it (name the role)	Year-one cost (hours × loaded rate), illustrative
Implementation	Project lead (ambulatory informatics); physician informaticist; practice managers	**888 h · $89,760** (project lead at half time for two quarters — 520 h at $95; physician informaticist: specialty note-template and workflow configuration, 4 h/week for 26 weeks — 104 h at $200; practice managers: 12 sites × 10 h of rooming, consent and device-workflow setup — 120 h at $55; vendor-led configuration sessions that are really your hours: 12 specialties × 2 sessions × 3 h × 2 attendees — 144 h at $90)
Integration and interfaces	Interface analyst; security analyst (identity, single sign-on, mobile-device management)	**490 h · $39,150** (EHR integration: note delivery, encounter matching, order and problem-list behavior, testing — 160 h at $90; single sign-on and mobile-device management for 140 phones; identity review — 120 h at $110; device provisioning and replacement, the labor only: 140 devices × 1.5 h — 210 h at $55)
Training — initial and ongoing	Two trainers; every clinician; new-hire onboarding	**800 h · $100,800** — 540 clinician-hours and 260 trainer or support hours (initial training: 140 clinicians × 2.5 h of their own time — 350 h at $150; trainers: curriculum, 24 sessions, at-the-elbow support in weeks 1–2 of each wave — 240 h at $70; post-go-live remediation in months 2–3: 140 clinicians × 1 h — 140 h at $150; ongoing onboarding: about 20 new hires a year × 2.5 h = 50 clinician-hours, plus about 20 trainer-hours — 70 h priced together at $150 as a pooled planning allowance)

The burden line (beyond the invoice)	Who carries it (name the role)	Year-one cost (hours × loaded rate), illustrative
Security, legal, and governance review	Security analyst; privacy officer; in-house counsel; the AI governance committee	**190 h · $28,100** (security assessment of the vendor and of the audio and transcript data flows — 40 h at $110; business associate agreement; contract terms on data use, model training and retention — 30 h at $150; privacy: patient notice, consent and opt-out workflow, signage and scripts, state-law check — 60 h at $120; AI governance committee: eight members × six meetings × 1 h, plus preparation — 60 h at $200)
Exception handling at steady state	Clinic coordinators; nurse informaticist; physician informaticist	**936 h · $77,480** (failed captures, misattributed speakers, opt-out encounters, late notes: coordinators at a quarter of one person — 520 h at $55; note-quality sampling: a risk-weighted sample of 40 signed notes a week read against the audio by the nurse informaticist, about 9 minutes each — 312 h at $90, derived below; physician informaticist: escalated quality findings, 2 h/week — 104 h at $200)
Support and escalation	Help desk; the analyst who owns the vendor relationship	**624 h · $41,600** (help desk tier one, a fifth of one person — 416 h at $55; vendor escalation owner: tickets, releases, outage calls, 4 h/week — 208 h at $90)
The champion's political capital	Two department chairs (their hours are priced here); the physician informaticist (her hours are priced in the implementation and exception rows; what she risks is recorded in words, not in hours)	**48 h · $12,000** (two chairs: 2 h a month each of persuasion, exception adjudication and defending the project — 48 h at $250)
Leadership attention	Executive sponsor (the chief operating officer); monthly steering	**96 h · $21,600** (executive sponsor: 2 h a month — 24 h at $300; monthly steering: six people × 1 h × 12 — 72 h at $200)
Total listed burden, year one	—	**4,072 h · $410,490**
Total: price + burden	—	**$672,000 + $410,490 = $1,082,490** (the listed burden is about 61 percent of the invoice under these assumptions; training is the largest cost category, and 540 of its 800 hours are the clinicians' own)

What the estimate measures — and what it does not. The $410,490 is a first-year planning estimate of the listed resource commitments, valued at loaded rates. It is not a claim that all of it becomes additional cash expenditure. Most of the hours are existing people doing new work: finance has to say which lines require new spending or backfill (the nurse informaticist role does not yet exist; the coordinator quarter and the help-desk fifth may or may not), which displace other work, and which are simply attention. It also has to say who carries each line: the 28 affiliated clinicians' own hours (about 108 of the 540 clinician-hours) are their practices' time, not Riverbend's, and who pays for their licenses and devices is a separate arrangement — the kind Chapter 17 says counsel designs before anything is offered, because the physician self-referral and anti-kickback laws may apply when a health system furnishes technology to physicians it does not employ. The sheet prices labor, not purchases: device purchases, any one-time vendor fees and any price escalator are left to the contract review. And the pilot priced below is part of these rows, not an addition to them; count the work once.

Three observations the filled sheet makes unavoidable:

1. **Training is the largest cost category, and most of it is the clinicians' own time — the time the business case called "savings."** The row is 800 workforce hours: 540 clinician-hours (initial training, remediation and onboarding) and 260 trainer or support hours. Initial training precedes use; remediation and new-hire onboarding run through the year, so the row is not a one-time cost paid before any benefit. The value story has to be measured, not assumed: the minutes a

clinician actually gets back after reviewing and correcting the draft (Chapter 17), against these hours and the rest of the right-hand column.

2. **Exception handling and support are permanent lines.** The sheet gives the exception queue an owner (a quarter of a coordinator) and a quality sampler (the nurse informaticist) for as long as the product runs; that is 1,560 hours a year that nobody budgeted in the demo.
3. **The ratio belongs to this case.** Here the listed burden is about 61 percent of the invoice. That number comes from these assumptions and is not a benchmark for ambient documentation or for enterprise tools generally; Chapter 6's observation that on many mid-sized tools the right-hand column "can rival or exceed the license in year one" is the reason to build the column, not a figure to compare against. In this scenario the integration line is comparatively light because the product is assumed to arrive pre-integrated through the record — the trusted lane of Chapter 17 — which is an assumption to verify for any real vendor, not a property of the category. An operator who sees her own number is not anti-product. She is the only person in the room pricing the whole purchase.

The three questions

What does the vendor say this costs the institution in year one, and is the answer a blank look? In this example the vendor's first answer was a blank look. Its second answer, two weeks later — a stipulated event in this scenario — was an itemized estimate of its own that priced every line except exception handling, which it had never priced for anyone. That answer did not earn the contract; the pilot and the gates below still had to be passed, and an itemized sheet is not evidence of implementation experience. What it did was make the remaining questions discussable — and give the reference calls (Instrument 15) a specific thing to ask about: who, at the vendor's other customers, carries the exception queue, and how many hours a week.

Who carries each line, by role? Every line above has a role. Two capacity gaps remain in the organization as it stands, and they need different remedies: the nurse informaticist who samples note quality does not yet exist — the role will be created and funded, which is itself a cost the invoice did not show; and the vendor-escalation owner is the interface analyst who already owns three other vendors — not a vacancy but an overload, which needs protected hours or a different assignment before go-live.

What does "free" actually consume? The vendor offered a free ninety-day pilot for twenty clinicians across three sites. Priced with this sheet, the pilot is not a twenty-clinician slice of the year; it is the front of the one-time work. The pilot subledger below (illustrative, like everything here) takes each line from the year-one rows above — the same hours, not new ones:

Pilot line (90 days, 20 clinicians, three sites)	Hours × rate, illustrative	Of the year-one row
Project lead, half time for the 13 pilot weeks	260 h × $95 = $24,700	of 520 h
Physician informaticist: core workflow and the templates for primary care and two specialties	40 h × $200 = $8,000	of 104 h
Practice managers, three pilot sites × 10 h	30 h × $55 = $1,650	of 120 h

Pilot line (90 days, 20 clinicians, three sites)	Hours × rate, illustrative	Of the year-one row
Vendor-led configuration sessions: three specialties × 2 sessions × 3 h × 2 attendees	36 h × $90 = $3,240	of 144 h
EHR integration, built once for every wave	160 h × $90 = $14,400	all 160 h
Single sign-on, mobile-device management and identity design, done once	80 h × $110 = $8,800	of 120 h
Device provisioning, 20 devices × 1.5 h	30 h × $55 = $1,650	of 210 h
Security, legal and governance review — all of it, before the first pilot encounter	190 h = $28,100	all 190 h
Trainers: curriculum, pilot sessions, at-the-elbow support in weeks 1–2	60 h × $70 = $4,200	of 240 h
Clinicians' own initial training, 20 × 2.5 h	50 h × $150 = $7,500	of 350 h
Exception queue during the pilot: a coordinator, 4 h a week × 13	52 h × $55 = $2,860	of 520 h
Pilot note sampling: 400 notes × about 9 minutes, by a borrowed quality nurse until the nurse informaticist exists	60 h × $90 = $5,400	of 312 h
Executive sponsor (6 h at $300) and steering (18 h at $200) during the pilot	24 h = $5,400	of 96 h

Pilot line (90 days, 20 clinicians, three sites)	Hours × rate, illustrative	Of the year-one row
The “free” pilot	**1,072 h · $115,900**	about 26 percent of the year-one hours and 28 percent of the year-one dollars

Of the three one-time lines — implementation, integration and the security, legal and governance review, 1,568 hours and $157,010 together — the pilot spends 826 hours and $90,540 before the organization has decided anything: about 53 percent of those hours and 58 percent of those dollars. “Free” was the most expensive line in the proposal until it was priced.

The sampling arithmetic. A percentage is not a staffing plan. The 312 hours above come from a defined sample: 40 signed notes a week, 2,080 a year, read against the audio at about 9 minutes each — 2,080 × 9 ÷ 60 = 312 hours. Against an estimated 492,800 ambient notes a year (140 clinicians × 16 ambient encounters a clinic day × 220 clinic days) that is about 0.4 percent, weighted toward new users, specialties with complex notes and any clinician the exception log flags. During the pilot the sampling is denser — 400 of an estimated 20,800 pilot notes, about 2 percent — because the pilot is the validation. Neither percentage demonstrates adequacy on its own; the governance committee sets the sampling plan, and revisits it quarterly against what the samples find.

Burden-shift check

Every automation must name who gets relief, who inherits work, and whether the enterprise delta is positive after verification, exceptions, reconciliation, training, monitoring, and renewal defense.

Who gets relief	Who inherits work	Is the enterprise delta positive?
Clinicians: after-hours documentation — measured as the clinician’s own time in the record after the clinic session, from the EHR’s time-in-system report, not the vendor’s minutes-saved metric	Medical assistants and front desk: patient notice and consent at rooming; the opt-out path. Coders: notes with a different structure. Coordinators: the exception queue. The privacy office: a new category of inquiry. The quality sampler: a new standing duty	**Not established by this worksheet.** The sheet prices the work; it does not measure the benefit. The charter below sets operating checkpoints — after-hours documentation time down by at least 30 percent from each clinician’s own baseline for at least 14 of the 20 pilot clinicians, and the exception queue inside its allowance — and those are conditions for continuing, not a finding that benefits exceed $1,082,490. The net enterprise value needs a separate estimate: the magnitude and timing of the relief after review and correction, the cash that is actually new, and the nonfinancial benefits (evenings returned, retention) that the CFO and the chief medical officer value together rather than pretend to have monetized. If either checkpoint fails, the delta is work moved, not removed, and the renewal meeting will be about that

The safety and governance steps the instruments require

The worksheet prices the purchase; it does not approve it. Two instruments sit beside it before go-live.

Instrument 6 — walk every safety word back to its mechanism. *Accountable:* the clinician reviews and edits the draft against the transcript and signs it; a stored draft is not a completed clinical note and the chart shows it as a draft; the institution’s unsigned-documentation policy applies to ambient drafts — in this example a draft older than 24 hours appears on the clinic’s unsigned list with notice to the clinician. *Transparent:* the audio and transcript retention policy is written down (how long, where, who can see them), and a note can be reconstructed against its transcript after the fact. *Monitored:* the note-sampling program described above, with its thresholds (misattributed speaker affecting clinical content, omitted finding, invented history), a dashboard, and a named person — the physician informaticist — who receives the alert when a threshold is crossed; the privacy officer receives data-flow alerts. *Governed:* the AI governance committee holds the controls — who can be added, what can be turned off, how a finding is escalated — and can defend them in front of the board; the committee, the

privacy officer or the physician informaticist can pause the tool for a clinician, a site or the whole pilot on a safety, privacy or material quality failure the day it is found, not at the day-90 meeting. *Compliant:* the business associate agreement, minimum-necessary data flows, the breach obligations, the patient-notice and consent workflow under state law, and the product's regulatory status, all dated before the pilot; and, because 28 of the clinicians are not employed, the arrangement for their licenses and devices reviewed by counsel before anything is offered. *Validated:* a local pre-pilot evaluation before the first patient-facing encounter — a defined set of encounters in which the clinicians review the drafts but the drafts are not filed to the chart, each draft read against the audio by the physician informaticist and a quality nurse, results in front of the committee; then the pilot sampling at about 2 percent of pilot notes; then the wave-two decision with the numbers in hand, on this population.

Instrument 8 — the pilot charter, six lines, before anything starts. Every value below is illustrative, chosen for the example; a real institution sets its own. If any cell were blank — an owner unnamed, a threshold unagreed, a stop condition undefined — the pilot would be on HOLD.

Charter line	Entered (illustrative)
Named owner — institution side	Dr. Imani Osei, physician informaticist (clinical owner); Daniel Reyes, project lead (operational owner); Lauren Whitfield, chief operating officer (executive sponsor; the decision)
Named owner — vendor side	the vendor's implementation lead, by name, with a named successor — both recorded in the charter
Scope and end date	20 clinicians (14 primary care, 6 across two specialties) at three sites; pilot day 1 is the Monday after the pre-pilot evaluation passes; 90 days; end date on the calendar; the day-90 decision meeting booked now
Success criteria — including burden criteria — agreed in writing before go-live	(1) After-hours documentation time: baseline = each clinician's own median minutes per clinic day in the record after the session, from the EHR time-in-system report over the eight weeks before the pilot (cohort median 62 minutes); target = at least a 30 percent reduction from the clinician's own baseline for at least 14 of the 20 clinicians, measured over pilot weeks 9–13. (2) Note quality: 400 sampled signed notes over the pilot (at least ten per clinician in weeks 1–2, then risk-weighted); material errors in the signed note — omitted finding, invented history, misattributed speaker affecting clinical content — in no more than 1 percent of sampled notes; any invented-history finding in a signed note reviewed by the physician informaticist within one business day; two for one clinician in a month pauses that clinician pending retraining. (3) Exception queue: coordinator time at or under 4 hours a week averaged over weeks 5–13 (the subledger's allowance; early exception rates run higher than steady state — the full-scale allowance of a quarter-time coordinator, 10 hours a week for 140 clinicians, is the measure at the checkpoint four months after wave one). (4) Patient opt-outs honored the same day, 100 percent of the opt-out log, reviewed weekly by the privacy office. (5) No ambient draft older than 24 hours without notice to the clinician; none older than 72 hours. (6) Stop conditions, effective immediately: any privacy or security incident involving audio or transcripts; three invented-history findings in signed notes across the cohort in any 30 days; any vendor use of the data outside the agreement
Measurement plan finance has blessed in advance	the EHR time-in-system report, pulled by finance's own analyst (the vendor's in-app minutes-saved figure is reported but is not the measure); the sampling log; the exception log; the opt-out log; assembled monthly by the project lead for steering and for the governance committee
Decision path: what converts the pilot, what kills it, and who makes that call	Day 90, on the calendar now: all criteria met → wave one (the remaining 120 clinicians in two waves), with a checkpoint four months after wave one on the same measures at scale and the renewal meeting in month ten, before the twelve-month term ends; any burden criterion failed → no wave one; re-price with the vendor or stop; the chief operating officer decides, with finance and the physician informaticist in the room. Safety, privacy and material quality failures do not wait for day 90: they pause the tool the day they are found (Instrument 6, *Governed*)

What this example teaches

The invoice was $672,000; the listed first-year price plus burden was $1,082,490. That arithmetic did not decide whether to buy, and it did not establish that buying pays. It made the purchase honest: the right-hand column named the two capacity gaps, priced the "free" pilot, separated the hours the institution commits from the cash they may or may not become, and

attached owners, measures and a date to the renewal meeting — a scheduled courtroom with the evidence already on the calendar.

These instruments are aids to structured thinking based on the author's operating experience. They are not legal, clinical, compliance, investment, accounting, tax, or consulting advice; they do not establish a standard of care or guarantee diligence or outcomes. Use them with qualified professionals and your own judgment. This worked example is a hypothetical composite; all names, organizations and figures are illustrative.